

Exercise Sheet for Maths Material Covered in Week 8 (Gaussian Elimination) + (Introduction to Vector Spaces)

Note that you will be given feedback for question (2).

- (1) Solve the following system of four linear equations in four variables x_1, x_2, x_3, x_4 over the set of rational numbers:

$$x_1 + 5x_2 - 2x_3 + 3x_4 = -11$$

$$3x_1 - 2x_2 + 7x_3 + x_4 = 5$$

$$-2x_1 - x_2 - x_3 - 2x_4 = 0$$

$$5x_1 + 3x_2 + 4x_3 - x_4 = 13$$

Show all the intermediate steps (not just the final values obtained for x_1, x_2, x_3, x_4).

Comment: For practice, solve first using the “standard” elimination method and then also using the compact way (using a matrix). Once you obtain values for x_1, x_2, x_3, x_4 plug them back into each of the four equations to verify that your answer is correct.

- (2) **(feedback)** On Friday I bought two candy bars, a croissant and three coffees for 13.5 pounds. On Saturday, I bought three candy bars, five croissants and two coffees for 21.5 pounds. On Sunday (when I was very sleepy the whole day) I bought a candy bar, three croissants and six coffees for 26.5 pounds. Tomorrow, I wish to purchase four candy bars, two croissants and one coffee. How much will this cost me?

Comment: Show your working, i.e., don't just write down the final answer.

- (3) Let V be a vector space over a field F . Show each of the following:

- Additive identity $\vec{0}$ is unique
- $0\vec{v} = \vec{0}$
- $s\vec{0} = \vec{0}$
- $(-1)\vec{v} = \overrightarrow{-v}$
- $s\vec{v} = \vec{0}$ implies $s = 0$ or $\vec{v} = \vec{0}$

Comment: To prove the above, you can use the following:

- Any of the 8 conditions available due to V being a vector space over F
- Any of the conditions available due to F being a field

- (4) Verify that $\mathbb{Q}^2$ is a vector space over the field $\mathbb{Q}$ with the following definition of vector addition and scalar multiplication:

(a) Vector addition: If $\vec{v}_1 = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$ then $\vec{v}_1 + \vec{v}_2$ is defined to be $\begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix}$

(b) Scalar multiplication: $\vec{v} = \begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{Q}^2$ and $r \in \mathbb{Q}$ then $r\vec{v} = \begin{pmatrix} ra \\ rb \end{pmatrix}$

Comment: Each of the 8 conditions in the definition of vector space over a field needs to be verified. You can use any of the conditions available due to $\mathbb{Q}$ being a field.